

Integers

Number World Study Notes

Positive numbers, negative numbers, zero, and operations on the number line

Whole numbers alone cannot describe everything we measure — temperatures below zero, money owed, or floors below ground level all need negative numbers. Integers extend the whole numbers to include negatives, giving us a complete picture of numbers on both sides of zero.

Key Definitions

Integers (Z)

The set of all positive whole numbers, negative whole numbers, and zero: $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$. Integers have no fractional or decimal part.

Positive Integers

Integers greater than zero (1, 2, 3, ...), lying to the right of 0 on the number line.

Negative Integers

Integers less than zero (-1, -2, -3, ...), lying to the left of 0 on the number line.

Absolute Value $|a|$

The distance of an integer a from 0 on the number line, always non-negative. For example, $|-7| = 7$ and $|7| = 7$.

Ordering & Operation Rules

- On the number line, numbers increase from left to right; any number to the right is greater.
- Every negative integer is smaller than every positive integer, and smaller than 0.
- Same signs: add the absolute values and keep the common sign. E.g., $(-4) + (-5) = -9$.
- Different signs: subtract the smaller absolute value from the larger, keep the sign of the bigger one. E.g., $8 + (-3) = 5$.
- Subtracting an integer is the same as adding its opposite: $a - b = a + (-b)$.
- Additive inverse: $a + (-a) = 0$, so every integer has an opposite that cancels it out.

Worked Examples

Example 1: Evaluate $-7 + (-5)$.

Both numbers are negative (same sign).

Add absolute values: $7 + 5 = 12$.

Keep the negative sign.

Answer: -12

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Example 2: Evaluate $8 + (-3)$.

Different signs.

Subtract smaller absolute value from larger: $8 - 3 = 5$.

The sign of 8 (larger) is positive.

Answer: 5

Example 3: Evaluate $-10 - (-4)$.

Rewrite subtraction as adding the opposite: $-10 - (-4) = -10 + 4$.

Different signs: $10 - 4 = 6$, keep sign of larger absolute value (-10 is negative).

Answer: -6

Example 4: Arrange -5, 3, -1, 0 in ascending order.

Ascending means smallest to largest.

Negative numbers are smaller than 0, and more negative is smaller.

Order: $-5 < -1 < 0 < 3$.

Answer: -5, -1, 0, 3

Practice Questions

- 1 Evaluate $-15 + 9$.
- 2 Evaluate $6 - (-6)$.
- 3 Which is greater: -8 or -3?
- 4 Arrange in ascending order: 4, -4, 0, -9, 7.
- 5 Find the additive inverse of -23.

Answer Key

- 1 -6
- 2 12
- 3 -3 (it is closer to zero, so it is the larger integer).
- 4 -9, -4, 0, 4, 7
- 5 23, since $-23 + 23 = 0$.